

NIYAMR AI Internship Assignment:

Technical Approach Report

This report compiles the evaluated approaches for each task of the assignment involving the Universal Credit Act 2025 PDF. Based on the document analysis and task requirements, we chose appropriate methods and will implement two complementary pipelines from Task 2 onwards: an API-assisted approach and a purely local, no-API approach.

Task 1: Text Extraction

- Document Nature: The Universal Credit Act 2025 is a digitally-created legal document, predominantly text-based, with structured sections, numbered clauses, schedules, and some tables.
- Approach: Use direct PDF text extraction tools such as PyMuPDF, pdfminer.six, or pypdf.
- Rationale:
 - Efficient extraction preserving structure (sections, subsections, tables).
 - OCR unnecessary, avoiding complexity and errors since the document is not scanned or image-based.
- Post-processing: Clean extracted text, segment logically, and convert to structured formats (JSON, etc.) for downstream tasks.

Task 2: Summarization

- Document Characteristics: Formal legal text with technical language, cross-references, and detailed clauses.
- Chosen Method: Hybrid summarization anchored on extractive methods.
 - Extractive Summarization: Select key sentences or paragraphs focusing on purpose, definitions, eligibility, obligations, enforcement. Techniques include TextRank, TF-IDF.
 - Light Abstractive Refinement: Use pre-trained NLP models (BART, T5, Pegasus) for lucid, concise bullet points maintaining legal accuracy.
- Implementation Steps:
 - Extract relevant sentences (e.g., using pytextrank or spaCy).
 - Categorize by logical headings based on keywords/section headers.
 - Preprocess for deduplication and simplification while preserving legal terms.
 - Refine abtractively with NLP models.
 - Final manual or automated validation for coherence and precision.
- Dual Approach:
 - Without API: Fully local extractive summarization with smaller fine-tuned models for light rewriting.
 - With API: Use LLM APIs (Gemini, GPT, etc.) for abstractive summarization phase to enhance readability and capture context.

Task 3: Key Legislative Section Extraction

- Potential Approaches:
 - Rule-Based Pattern Matching: Regex and keyword rules for legal section headings.
 - NER and Information Extraction Models: Use pre-trained or fine-tuned legal NLP models.
 - Layout & Semantic Analysis: Utilize PDF layout features combined with NLP.
 - Retrieval-Augmented Generation (RAG) or LLM Querying for context-aware extraction.
 - Hybrid: Combine methods for balanced accuracy and scalability.
- Dual Implementation Plan:
 - Without API:
 - Preprocess text, extract sections with regex and keyword rules.
 - Use NLP libraries (spaCy, NLTK) to detect entities and clause sentences.
 - Structure output as JSON.
 - With API:
 - Segment text and query LLM APIs with precise prompts.
 - Parse API responses to build JSON legal sections.
 - Validate results with cross-checking and prompt refinements.

Task 4: Rule Compliance Checks

- Approaches Considered:
 - Rule-Based Keyword & Pattern Matching.
 - Semantic Search with Sentence Embeddings.
 - LLM-Based Verification via direct querying.
 - Hybrid combining rule filtering and LLM validation.
- Without-API Approach :
 - Unsupervised keyword extraction (RAKE, TextRank, YAKE).
 - Semantic clustering or topic modeling to group rule-relevant text.
 - Legal NER models to automatically detect relevant entities.
 - Bootstrapping with seed terms for keyword expansion.
 - Determine rule satisfaction based on extracted evidence snippets.
- With API Approach:
 - Draft specific prompts for each compliance rule.
 - Query LLM APIs (Gemini, GPT, Claude) with or without context.
 - Extract, parse, and normalize answers into JSON status, supporting evidence, and confidence.
 - Post-process and validate outputs to ensure reliability.

Dual-Method Strategy Summary

- From Task 2 onwards, two parallel methods will be run:
 - Local, no-API pipeline emphasizing explainability, offline capabilities, and classical NLP.
 - API-assisted pipeline leveraging LLMs for semantic comprehension, abstraction, and enhanced accuracy.

- This strategy showcases skills with foundational NLP and cutting-edge AI, provides comparative insights, and presents fallback solutions.
- Both pipelines generate JSON outputs formatted per assignment requirements, enabling comparative evaluation.

Conclusion

This comprehensive approach balances accuracy, efficiency, cost, and technical versatility suited for the NIYAMR AI internship challenge. The dual focus on local and API-based methods reflects practical engineering wisdom and prepares for real-world constraints and innovation.